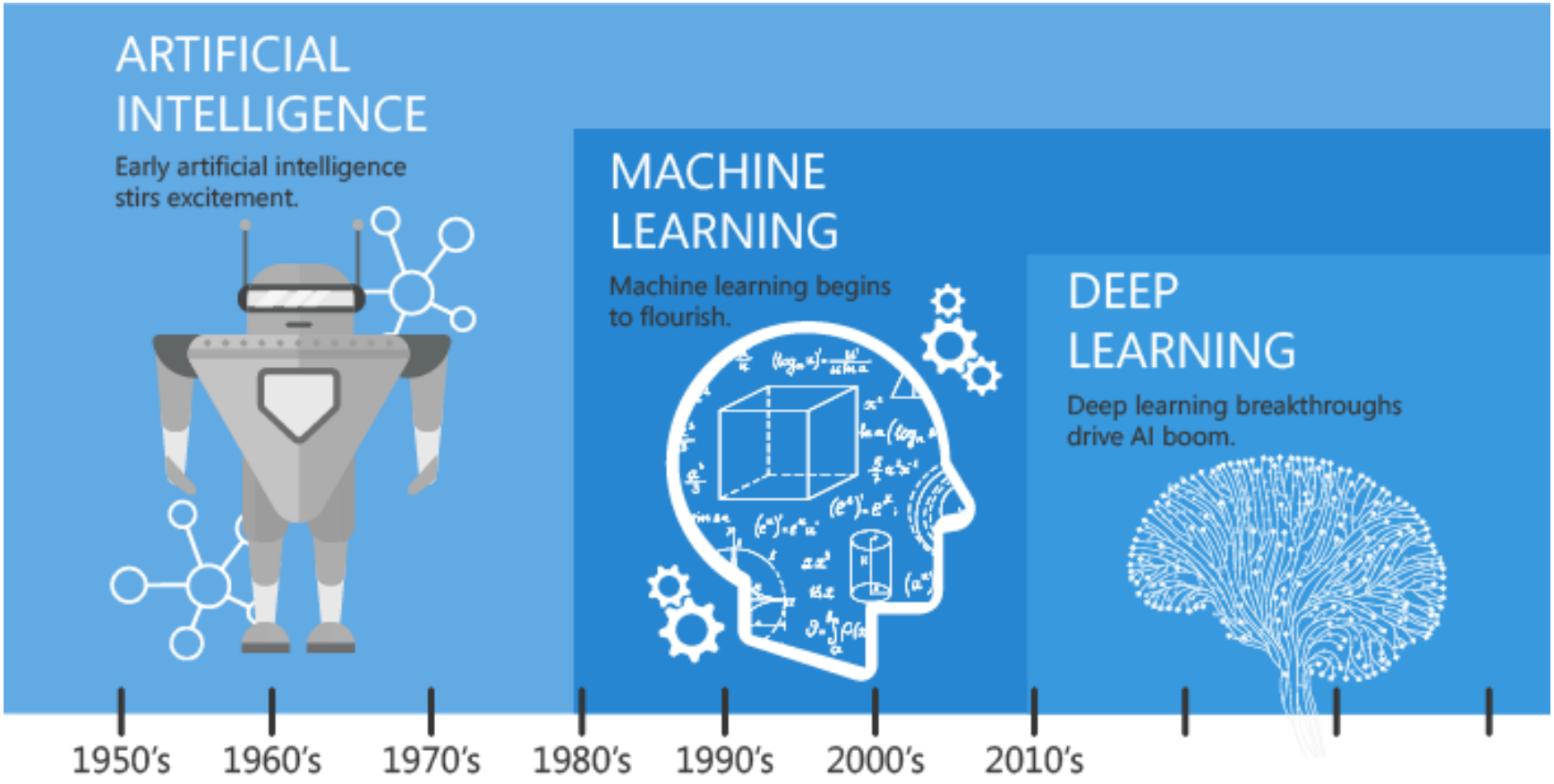


Deep Learning

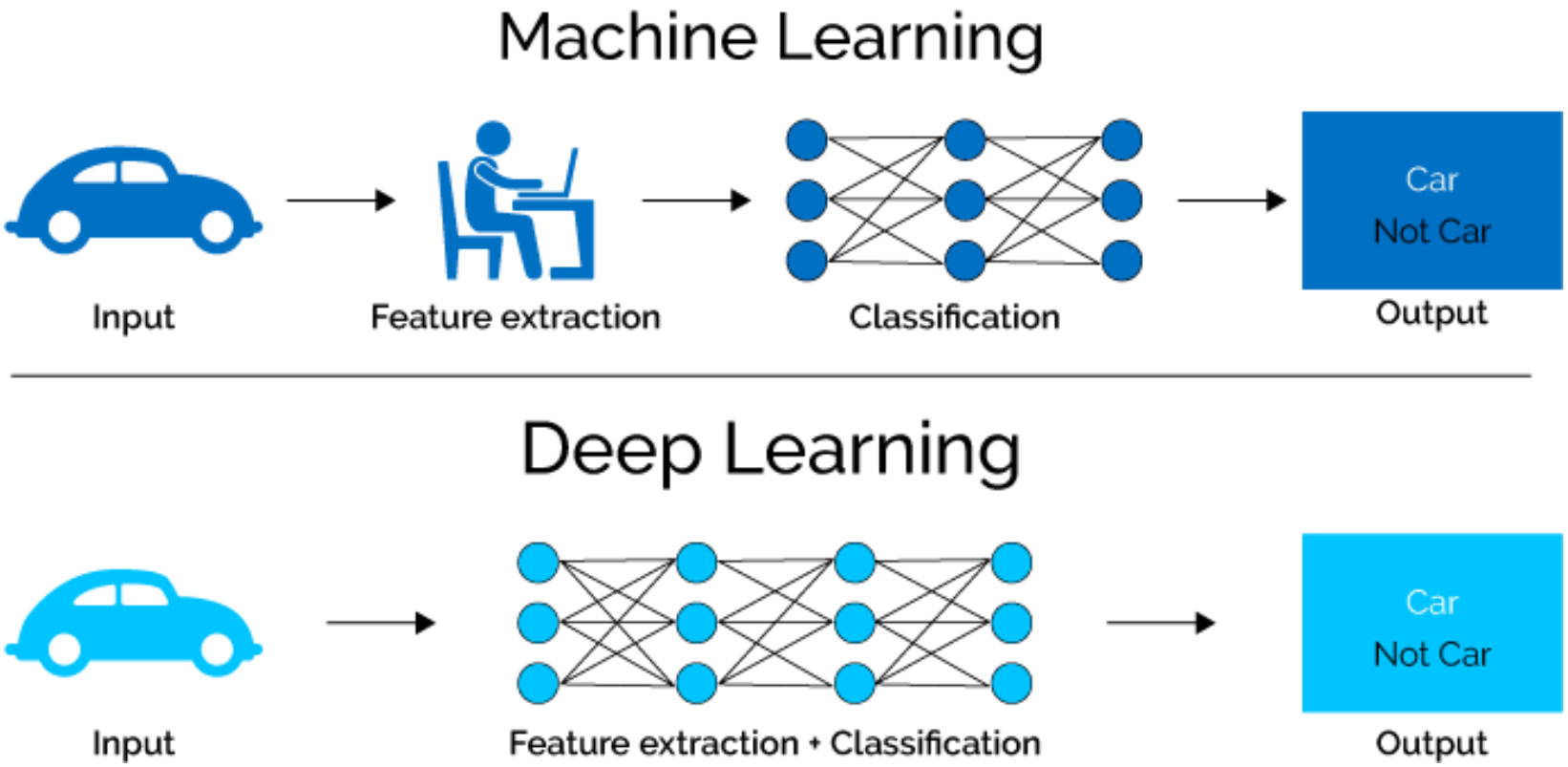
Prof. Hindriyanto D. Purnomo, Ph.D

Pengantar Deep Learning



http://www.smartcityindo.com/2019/11/difference-between-artificial.html

Pengantar Deep Learning



https://warstek.com/2018/02/06/deepmachinelearning/

Convolutional Neural Network

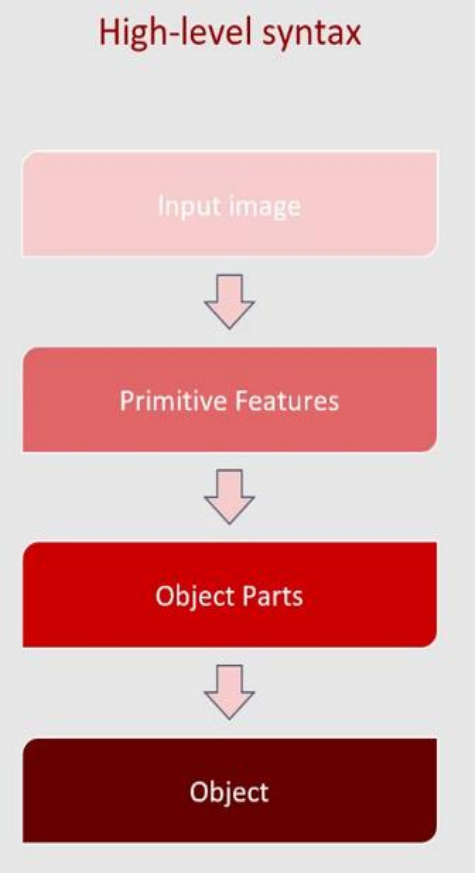
Convolution Neural Network

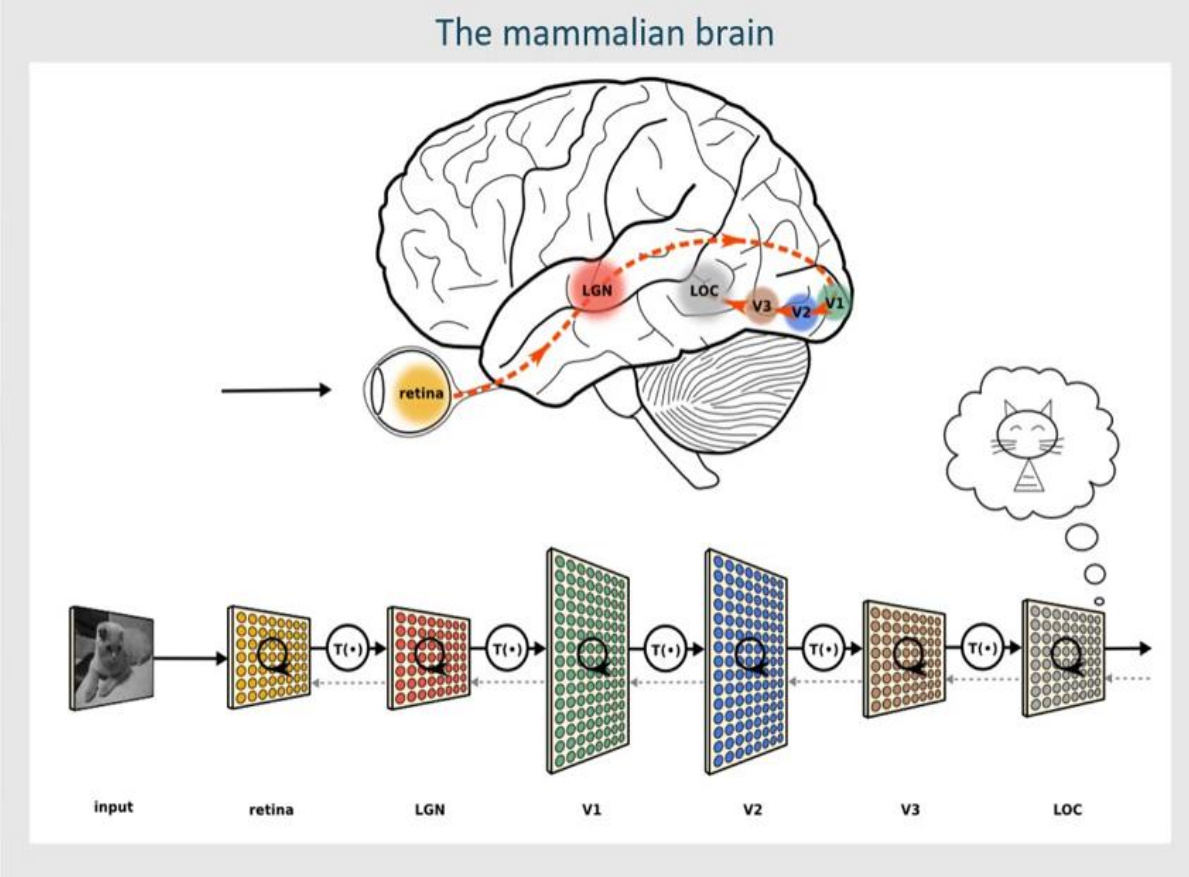
"How to create good representations of the visual world in a way it could be used to support recognition?"

Fitur utama:

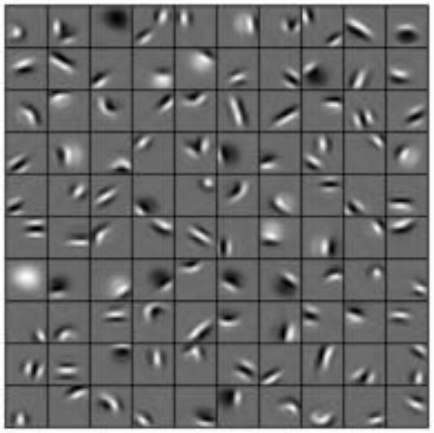
- **Mendeteksi dan mengklasifikasikan** objects kedalam kategori-kategori
- **Independen** terhadap pose, skala, iiluminasi, bentuk (*conformation*) and noise (clutter)

Convolution Neural Network

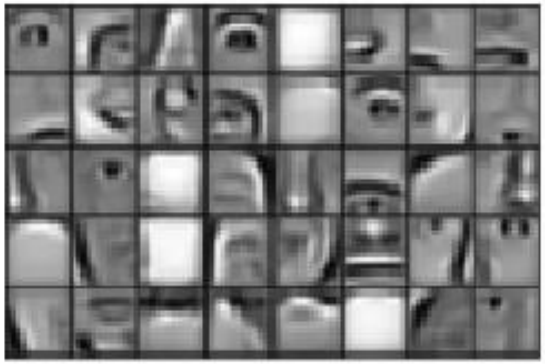




Convolution Neural Network



Fitur primitif

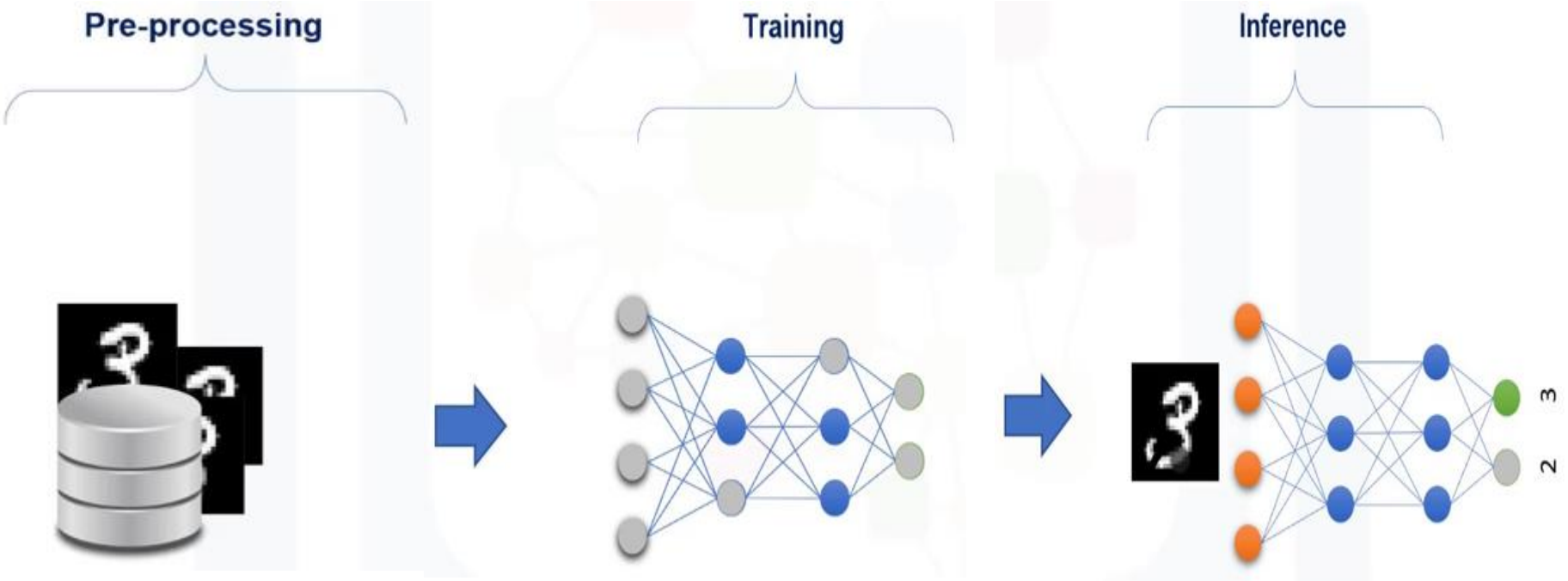


Bagian objek



Objek

Convolution Neural Network

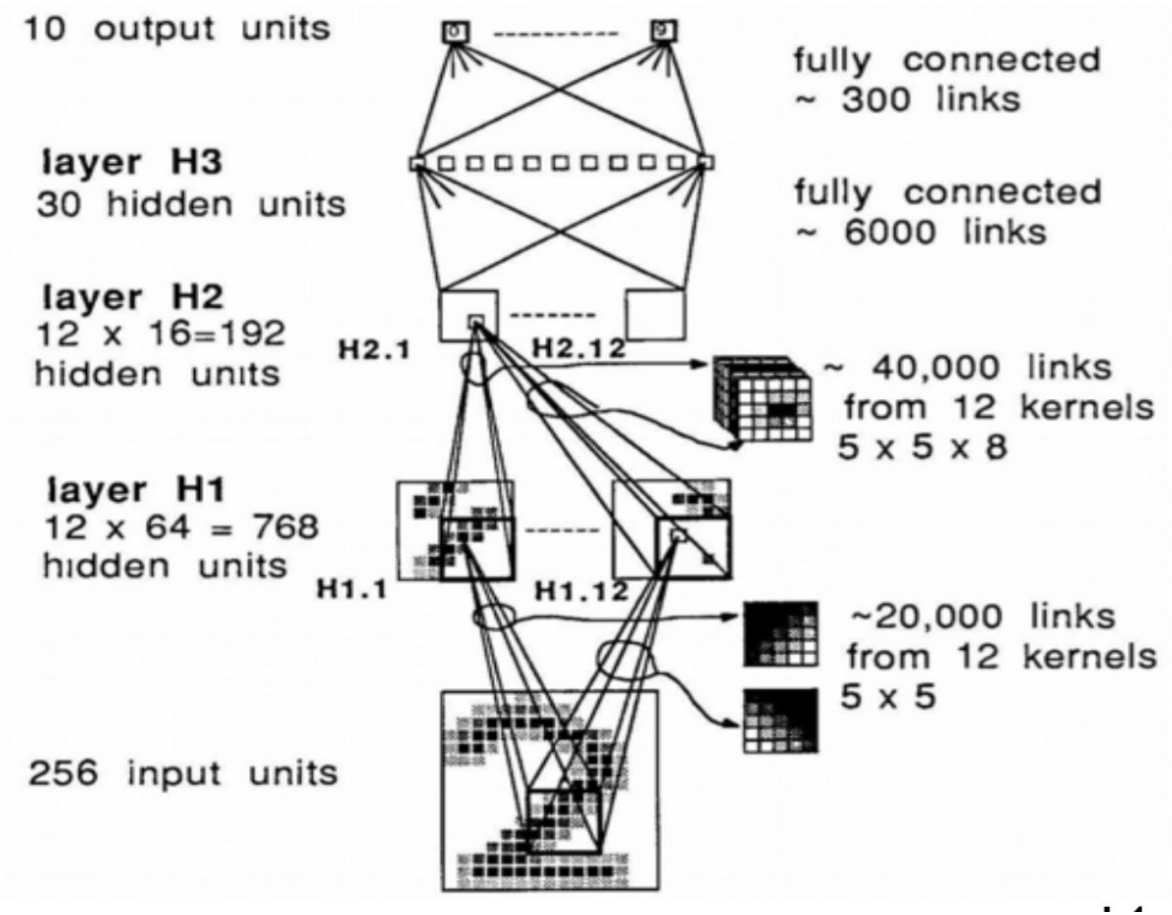


Convolution Neural Network

Fully connected Neural Network: setiap neuron akan terhubung dengan semua neuron di ayer sebelumnya

Convolutional Neural Network : setiap neuron hanya terhubung dengan sebagian kecil neuron di layer sebelumnya

Convolution Neural Network



http://yann.lecun.com/exdb/publis/pdf/lecun-89e.pdf

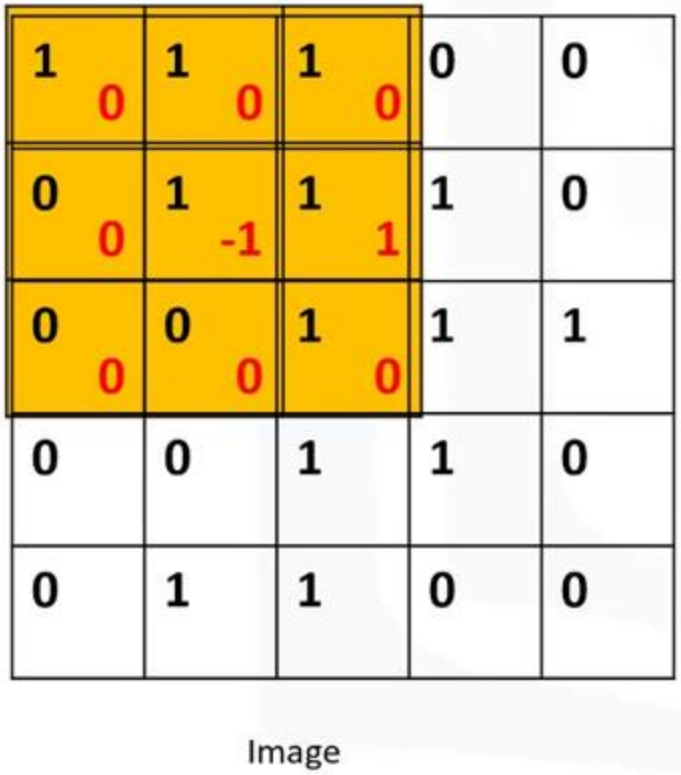
Convolution Neural Network

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Image

0	0	0
0	-1	1
0	0	0

Convolution Neural Network



Image

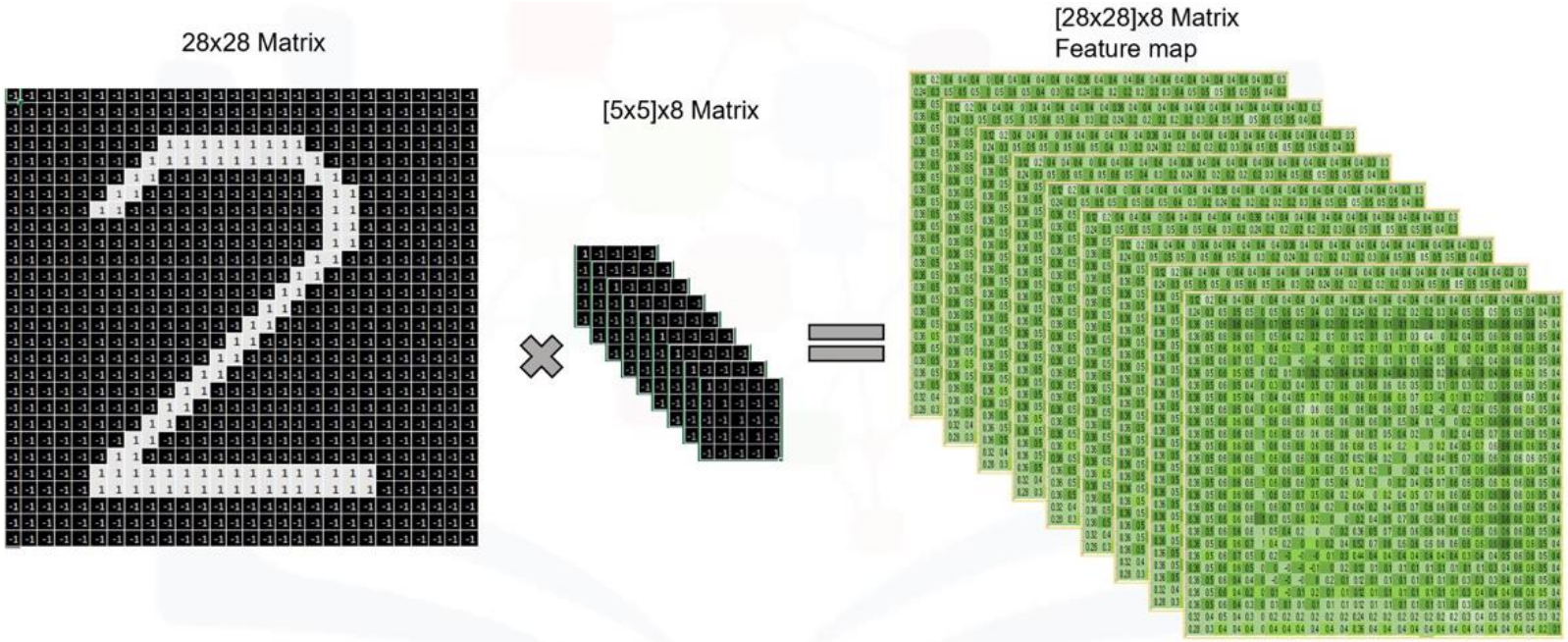
Convolution Neural Network



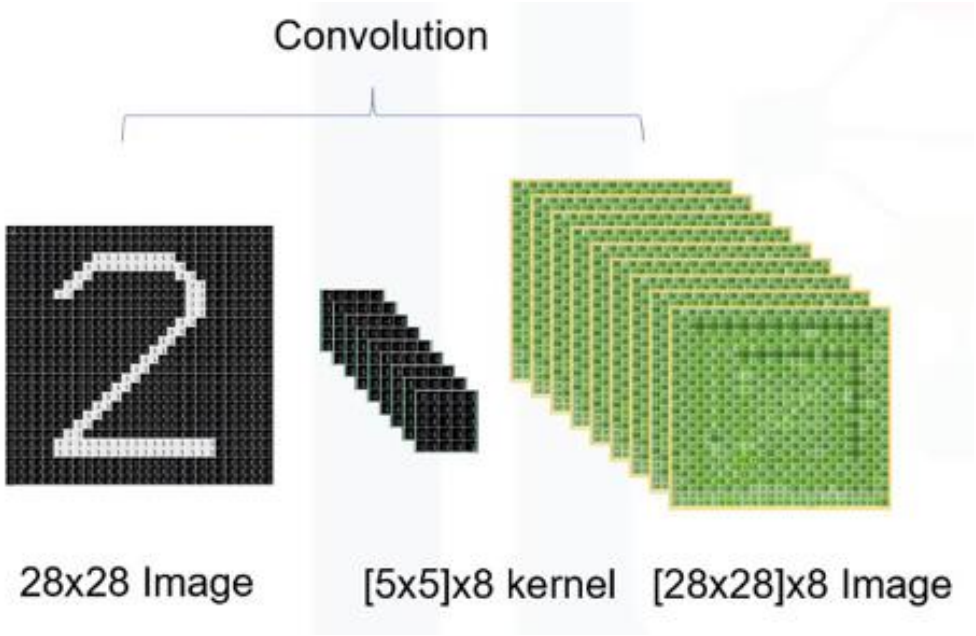
Convolution Neural Network



Convolution Neural Network

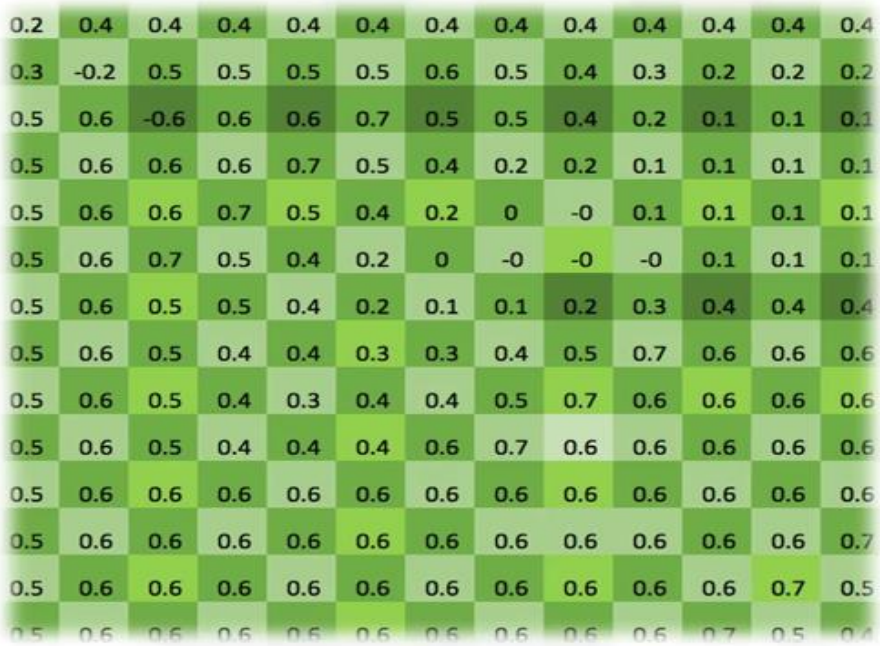


Convolution Neural Network

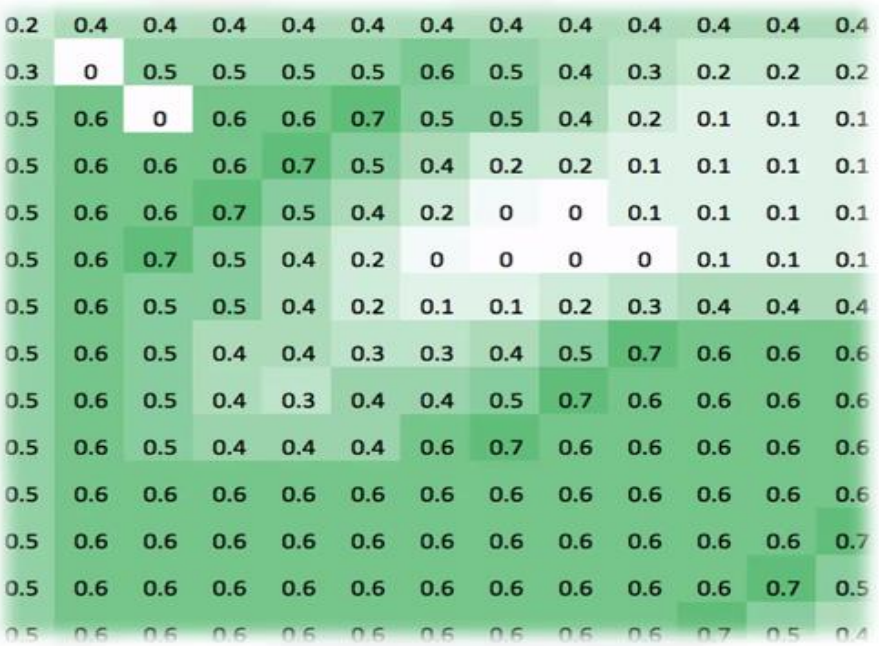


Convolution Neural Network

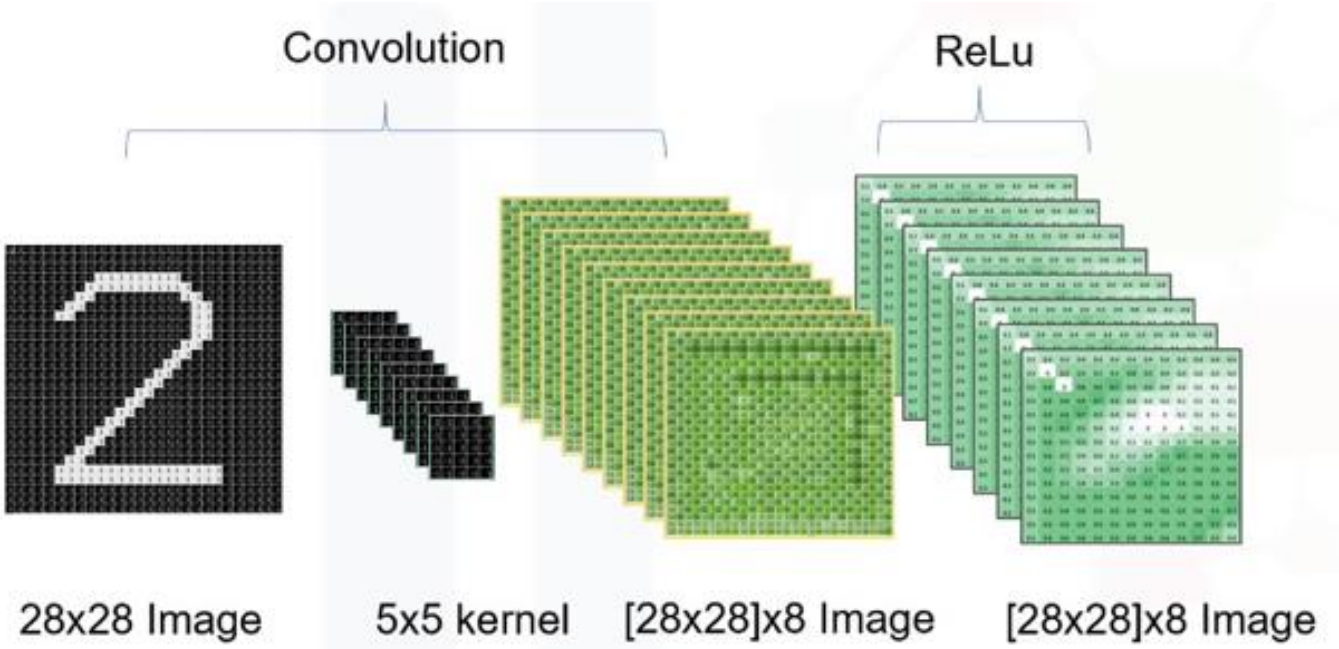
RELU



convolved1

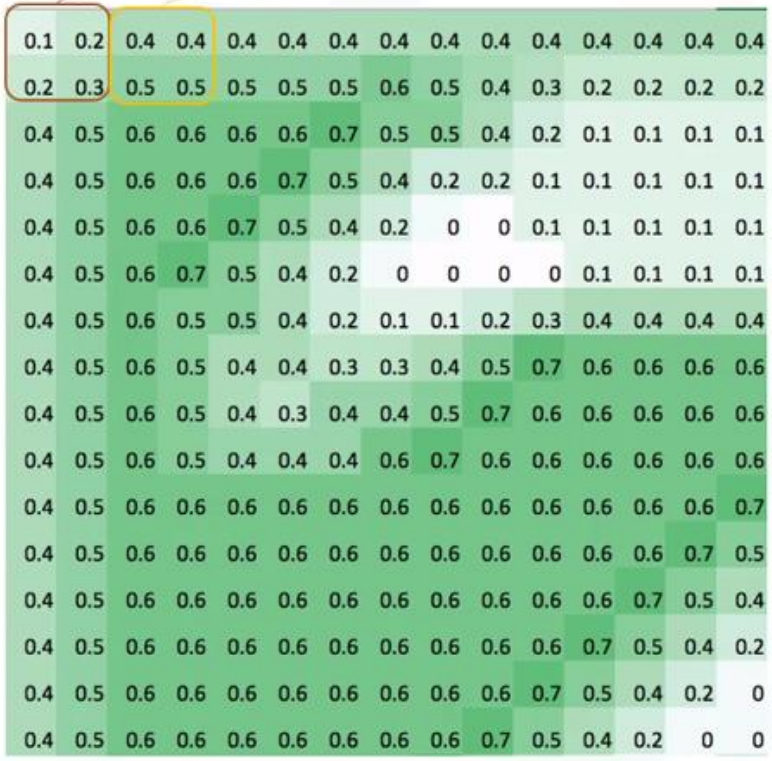


Convolution Neural Network



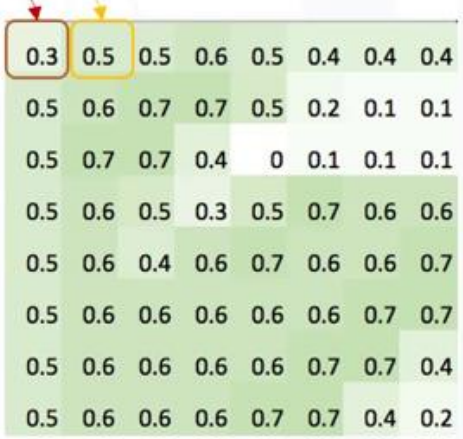
Convolution Neural Network

Max Pooling



0.1	0.2	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
0.2	0.3	0.5	0.5	0.5	0.5	0.5	0.6	0.5	0.4	0.3	0.2	0.2	0.2	0.2
0.4	0.5	0.6	0.6	0.6	0.6	0.7	0.5	0.5	0.4	0.2	0.1	0.1	0.1	0.1
0.4	0.5	0.6	0.6	0.6	0.7	0.5	0.4	0.2	0.2	0.1	0.1	0.1	0.1	0.1
0.4	0.5	0.6	0.6	0.7	0.5	0.4	0.2	0	0	0.1	0.1	0.1	0.1	0.1
0.4	0.5	0.6	0.7	0.5	0.4	0.2	0	0	0	0	0.1	0.1	0.1	0.1
0.4	0.5	0.6	0.5	0.5	0.4	0.2	0.1	0.1	0.2	0.3	0.4	0.4	0.4	0.4
0.4	0.5	0.6	0.5	0.4	0.4	0.3	0.3	0.4	0.5	0.7	0.6	0.6	0.6	0.6
0.4	0.5	0.6	0.5	0.4	0.3	0.4	0.4	0.5	0.7	0.6	0.6	0.6	0.6	0.6
0.4	0.5	0.6	0.5	0.4	0.4	0.4	0.6	0.7	0.6	0.6	0.6	0.6	0.6	0.6
0.4	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.7
0.4	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.5
0.4	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.5	0.4
0.4	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.5	0.4	0.2
0.4	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.5	0.4	0.2	0
0.4	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.5	0.4	0.2	0	0

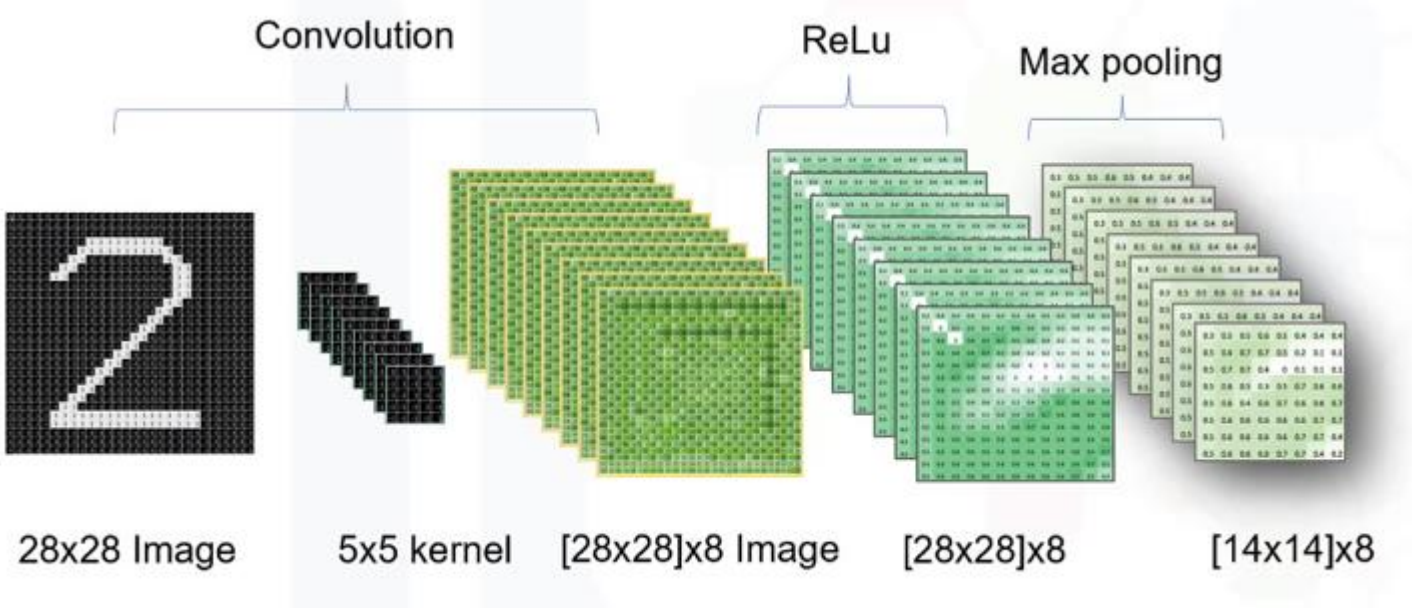
conv1_ReLu



0.3	0.5	0.5	0.6	0.5	0.4	0.4	0.4
0.5	0.6	0.7	0.7	0.5	0.2	0.1	0.1
0.5	0.7	0.7	0.4	0	0.1	0.1	0.1
0.5	0.6	0.5	0.3	0.5	0.7	0.6	0.6
0.5	0.6	0.4	0.6	0.7	0.6	0.6	0.7
0.5	0.6	0.6	0.6	0.6	0.6	0.7	0.7
0.5	0.6	0.6	0.6	0.6	0.7	0.7	0.4
0.5	0.6	0.6	0.6	0.7	0.7	0.4	0.2

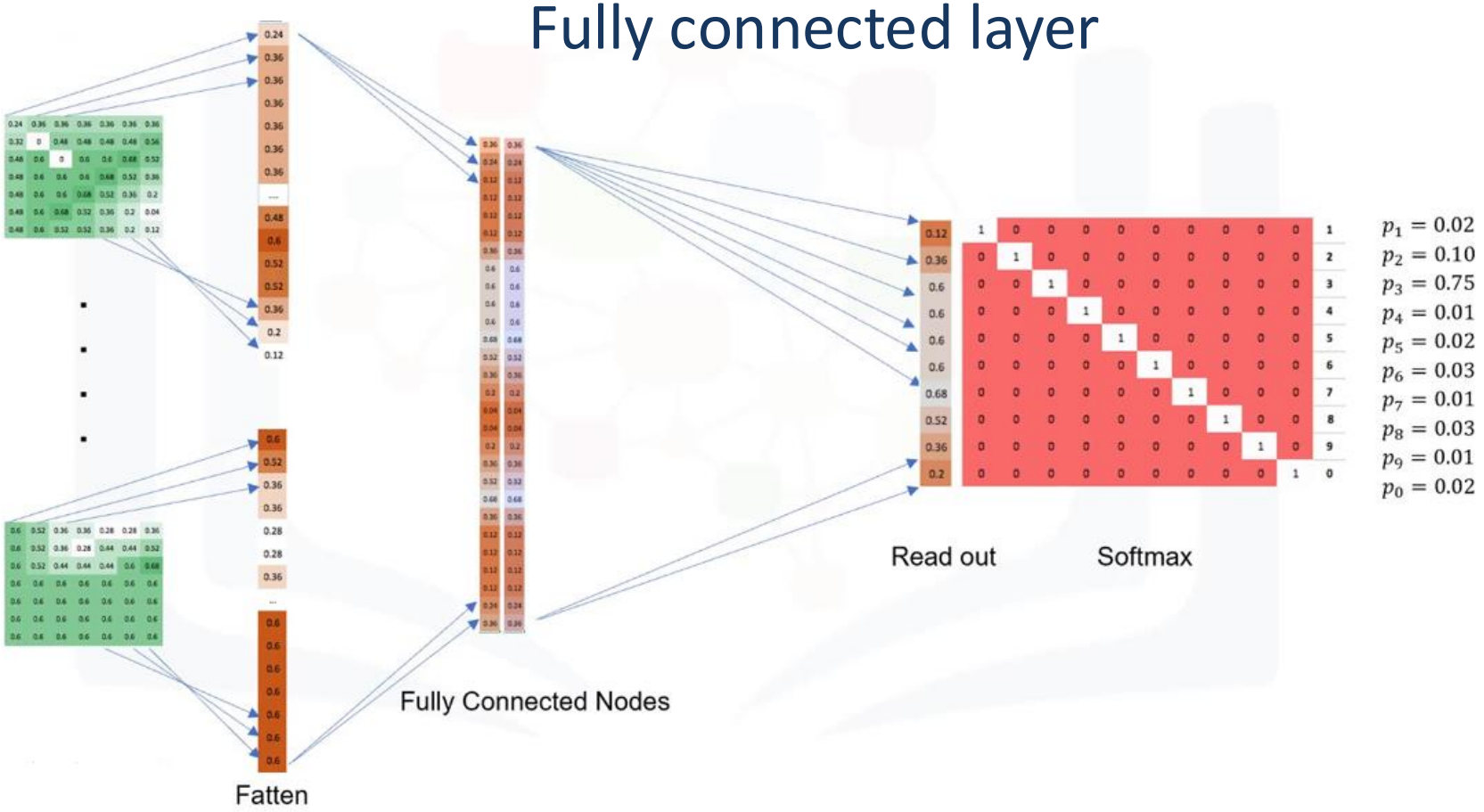
tf.nn.max_pool(conv1_ReLu, ksize=[1, 2, 2, 1], strides=[1, 2, 2, 1])

Convolution Neural Network



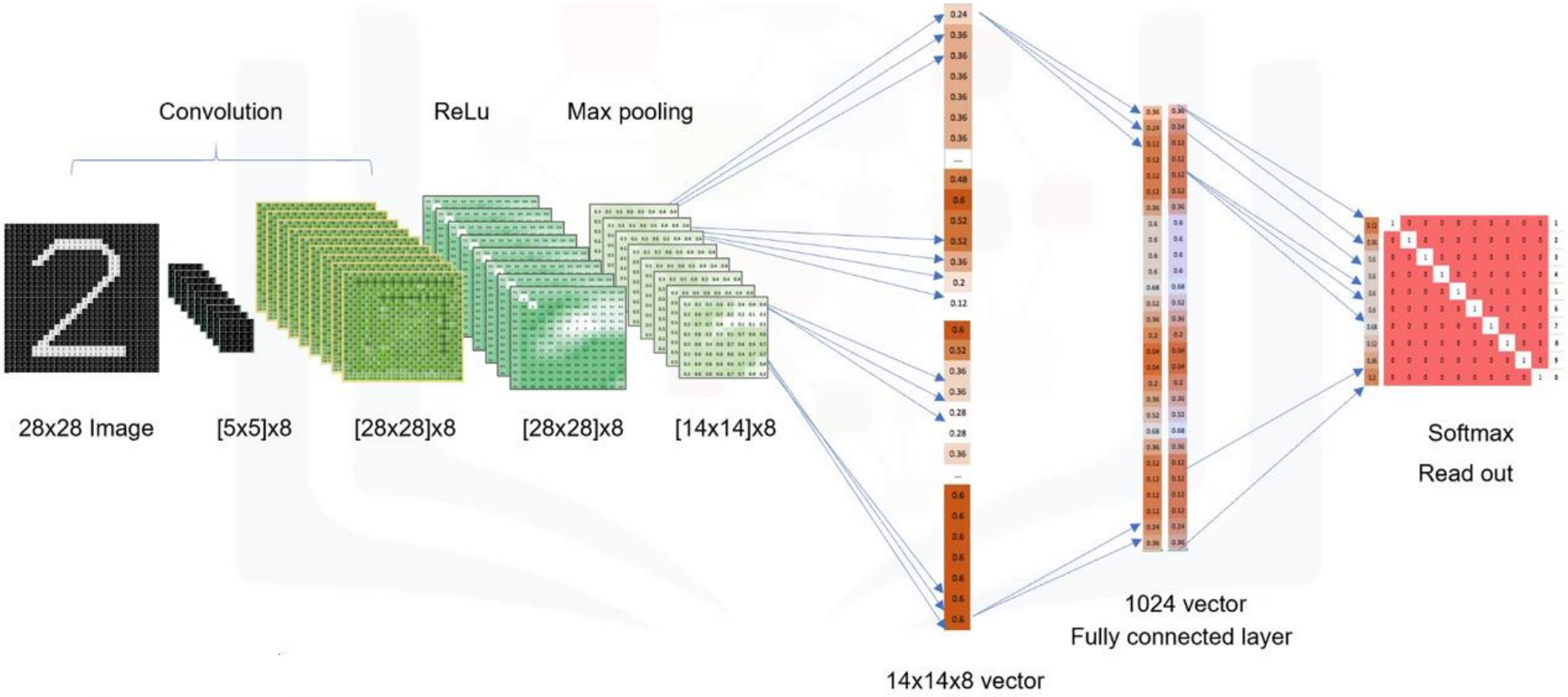
Convolution Neural Network

Fully connected layer

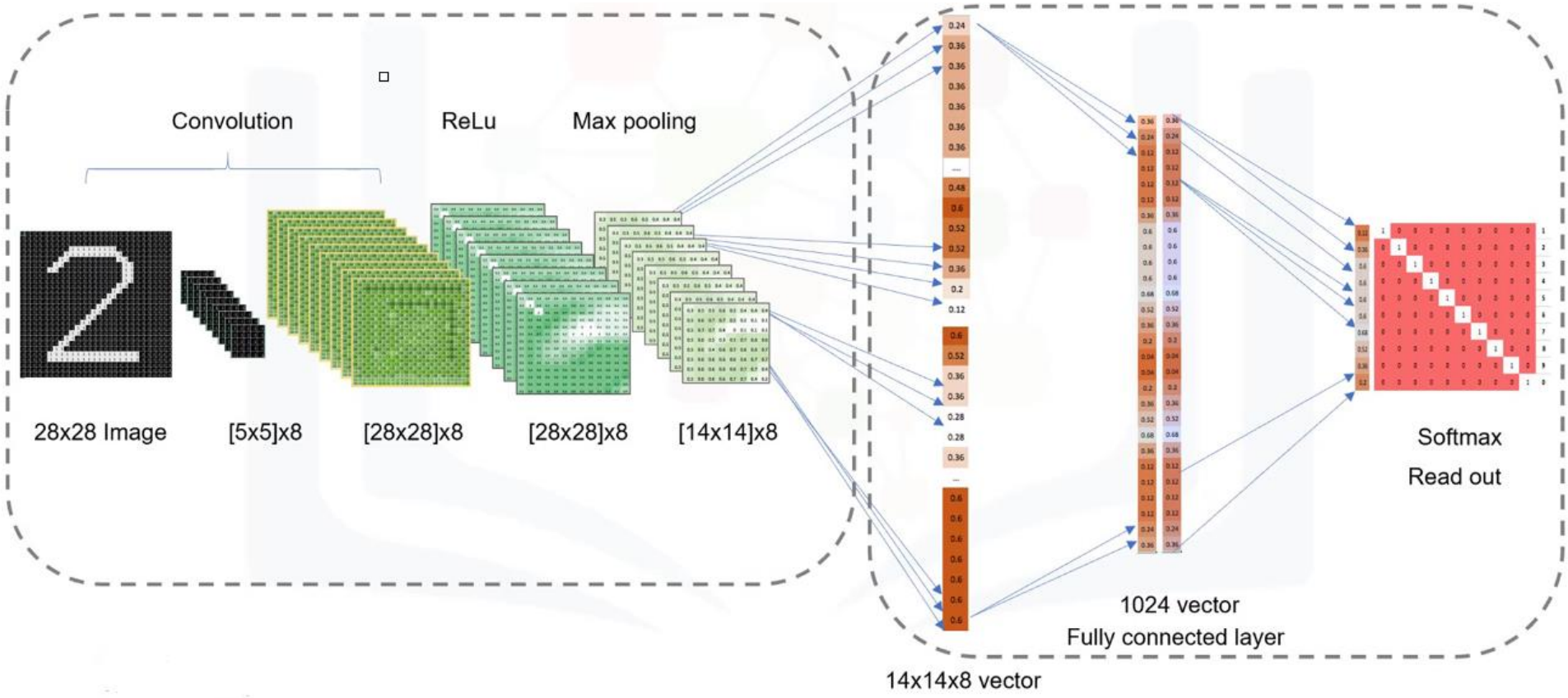


Convolution Neural Network

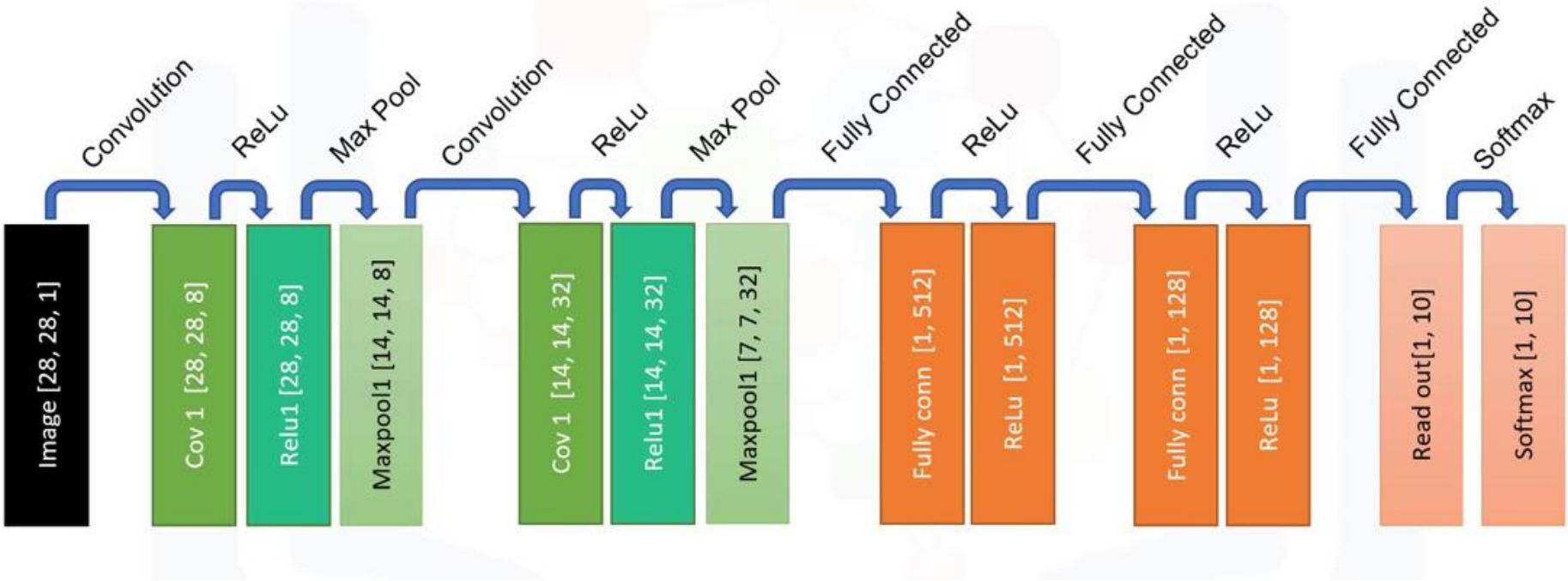
CNN Architecture



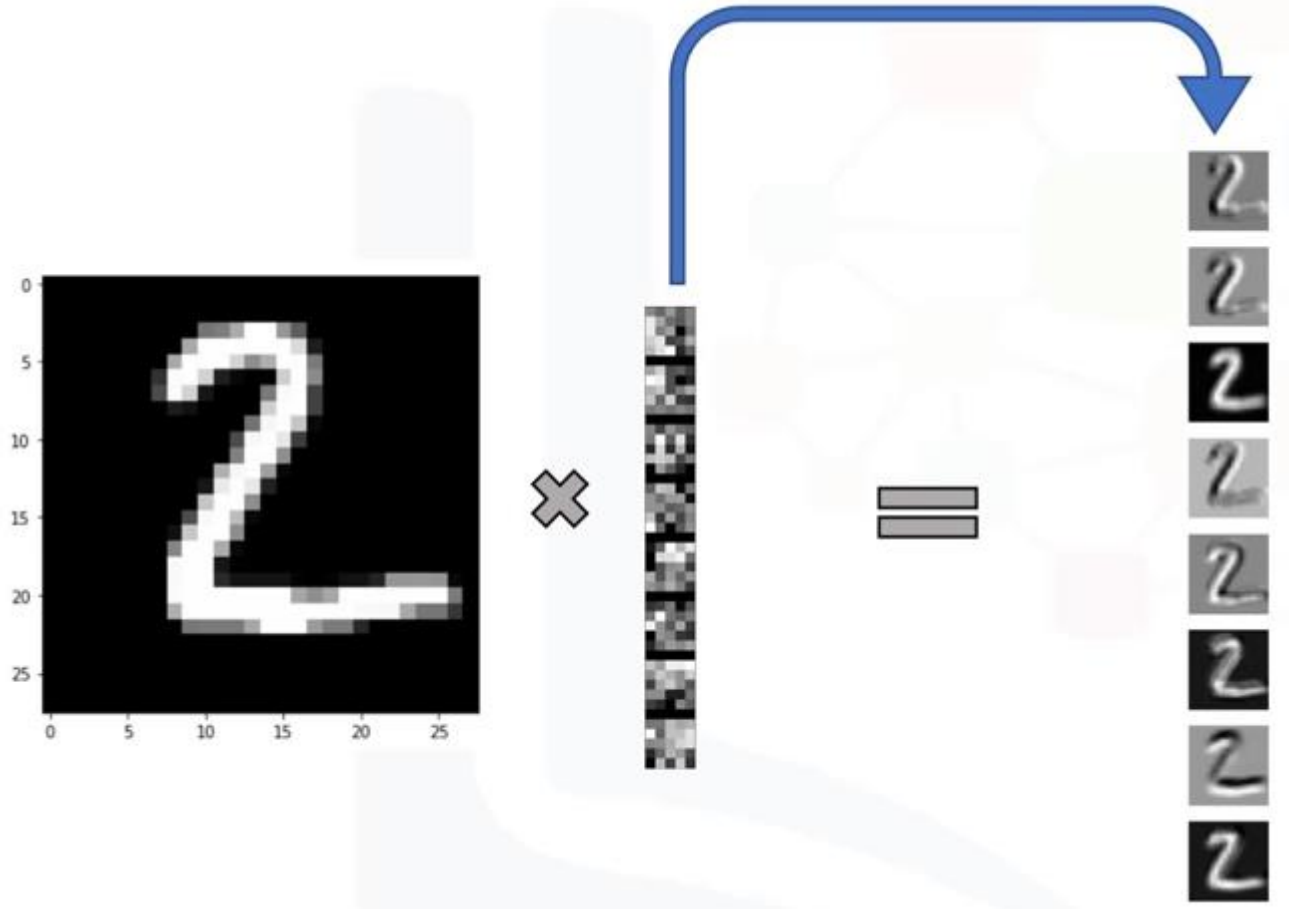
Convolution Neural Network



Convolution Neural Network



Convolution Neural Network

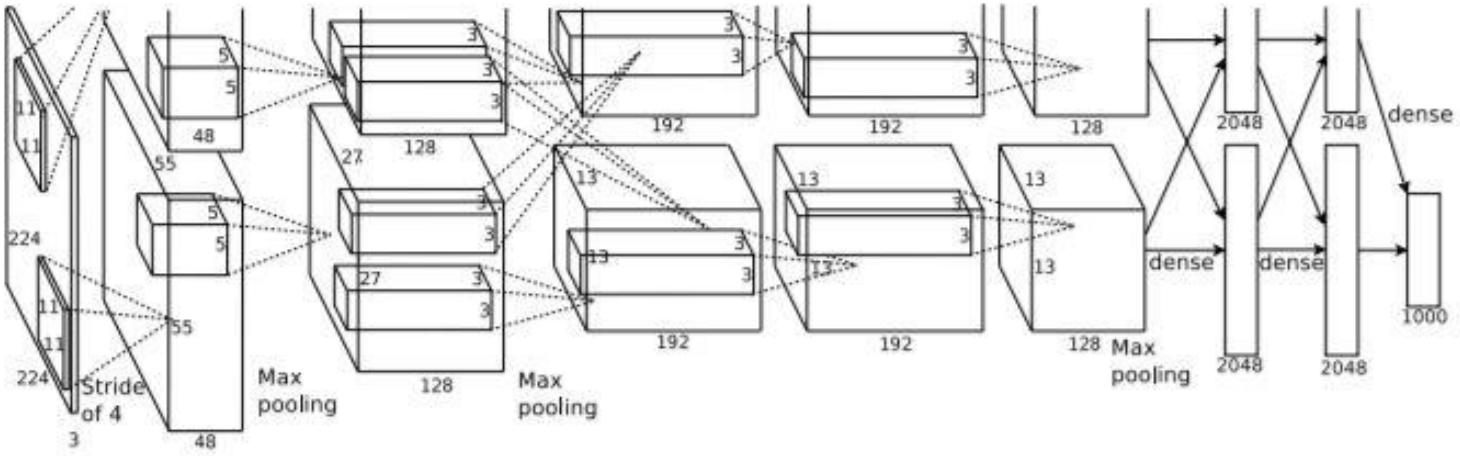


Contoh arsitektur CNN

1. Alexnet (2012) : pemenang Imagenet competition 2012

   5 convolution layer

   3 fully connected layer



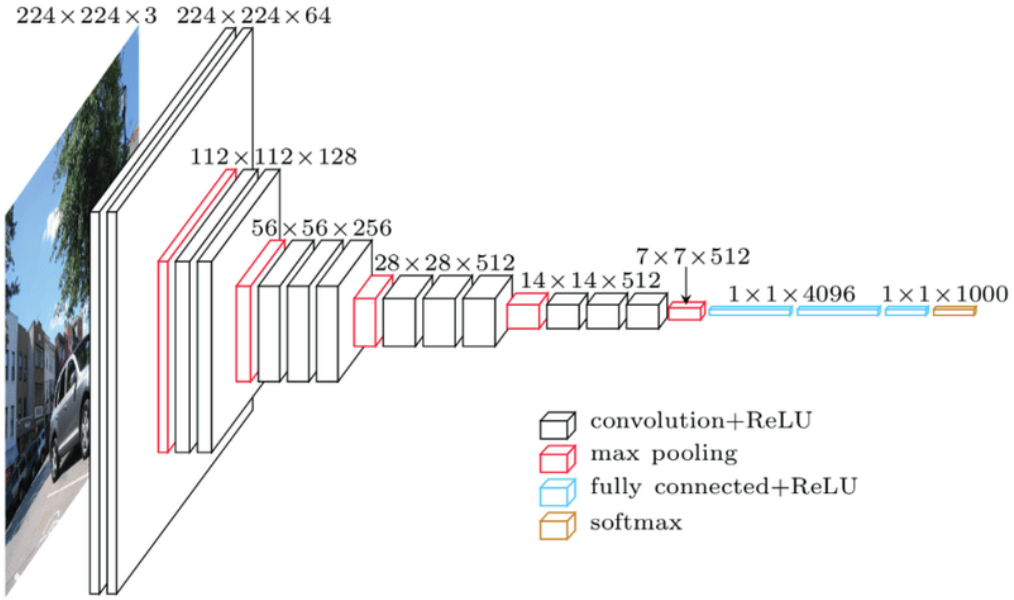
https://medium.com/ai-research-lab-kampala/alexnet-a-brief-review-14979ce7cc84

Contoh arsitektur CNN

2. VGGnet (2013)

   didasarkan pada Alexnet

   lebih sederhana

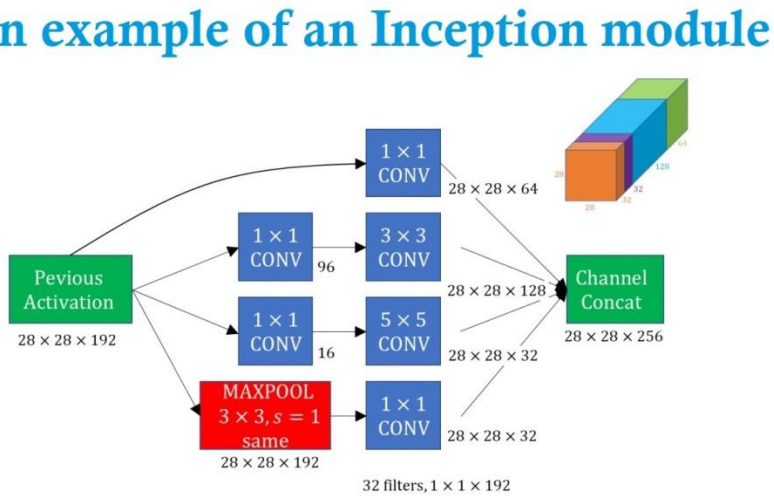


https://neurohive.io/en/popular-networks/vgg16/

Contoh arsitektur CNN

3. GoogleLeNet (2014):

   memperkenalkan konsep inception modul (beberapa koneksi dihilangkan untuk mengurangi computational cost dan overfitting)

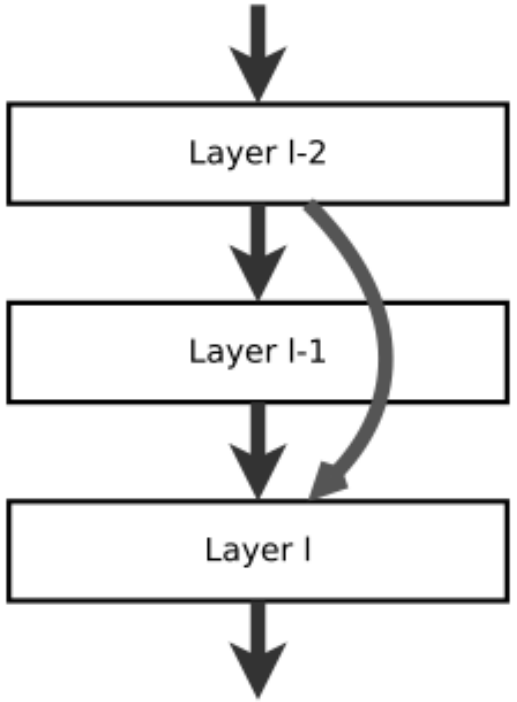


http://datahacker.rs/deep-learning-inception-network/

Contoh arsitektur CNN

4. Microsoft Resnet (2015)
   memungkinkan untuk
   membuat arsitektur dengan
   ratusan atau ribuan layer

   memperkenalkan konsep
   “identity shortcut connection”
   melakukan skip untuk satu
   atau lebih layer



https://en.wikipedia.org/wiki/Residual_neural_network

Terima kasih